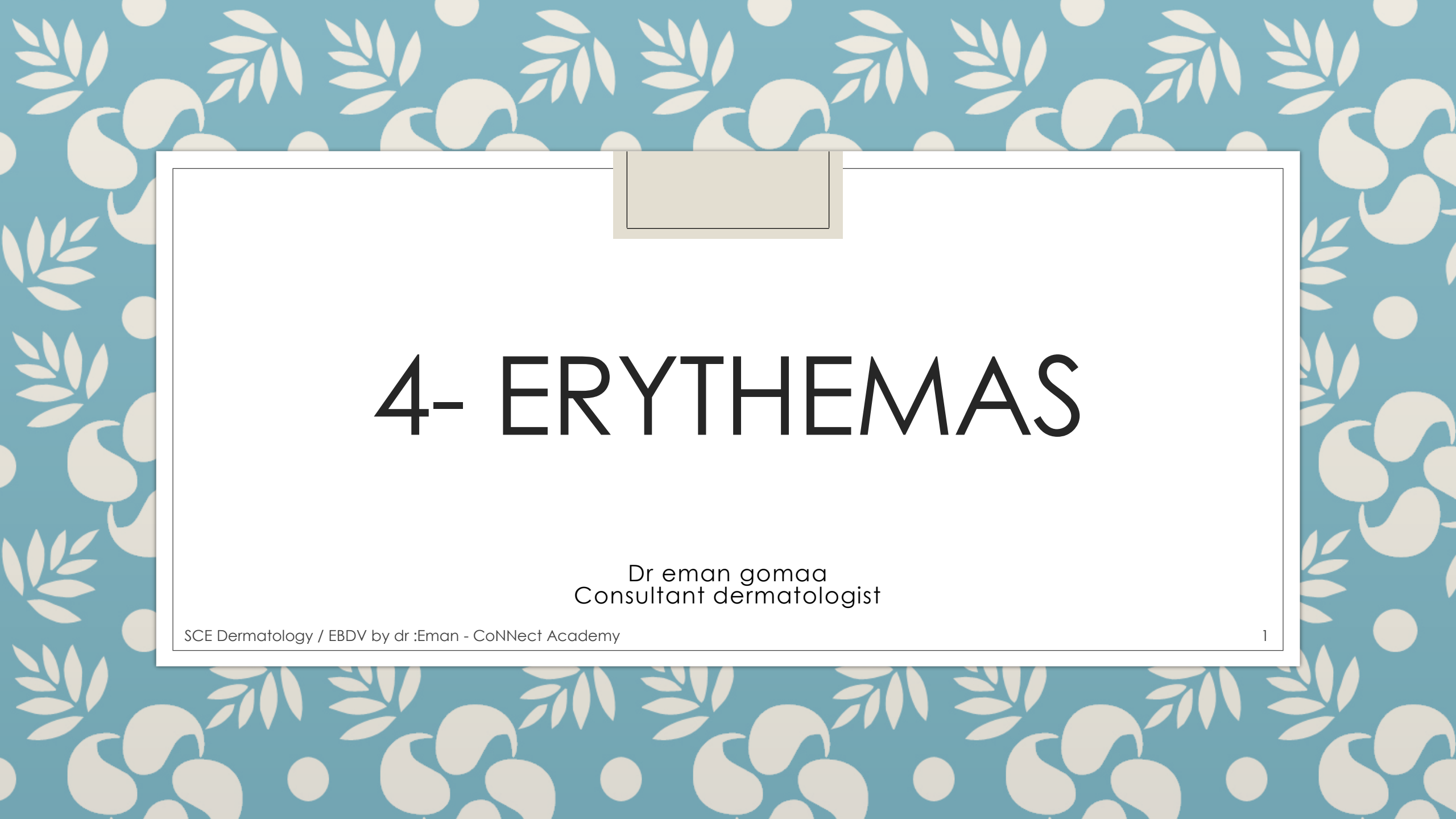




4- ERYTHEMAS

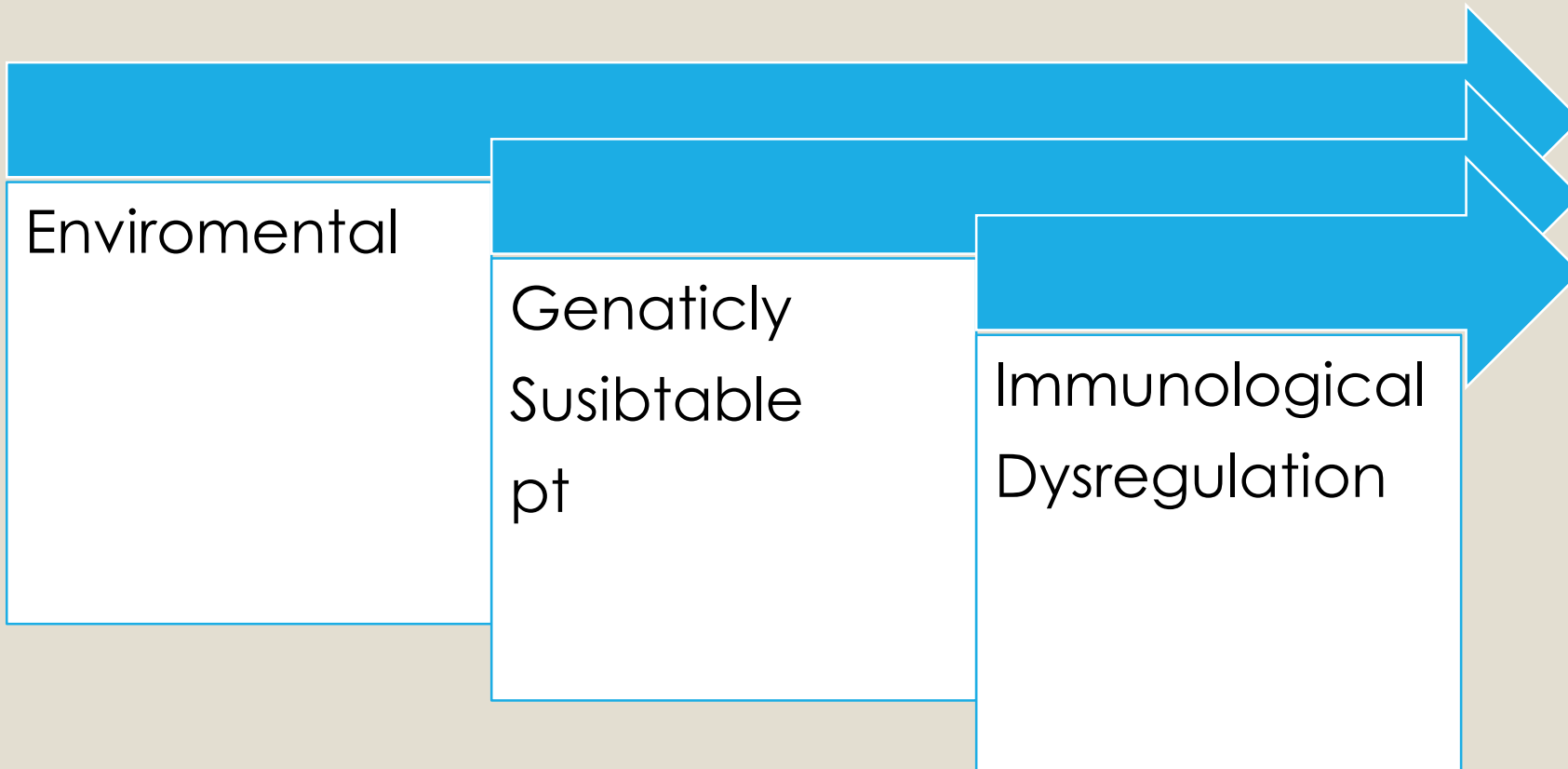
Dr eman gomaa
Consultant dermatologist

Eythema multiforme



Erythema multiforme (EM)

- acute, self-limited muco-cutaneous cytotoxic reaction to viral infection
- young adults slight male predominance.
- Target lesion
- Lesions appear in successive crops for a few days and fade in 1-2 weeks.
 - Lesions heal without sequelae



- **Genetic predisposition:**
 - An association with certain HLA types e.g.
- HLA-B15, -B35 and -DR53
- Precipitating factors
- Herpes simplex virus (HSV-1, HSV-2)
- Immunological reaction:
 - EM is a delayed hypersensitivity reaction. CD4 lymphocytes are found in the dermis and CD8 cells in the epidermis.

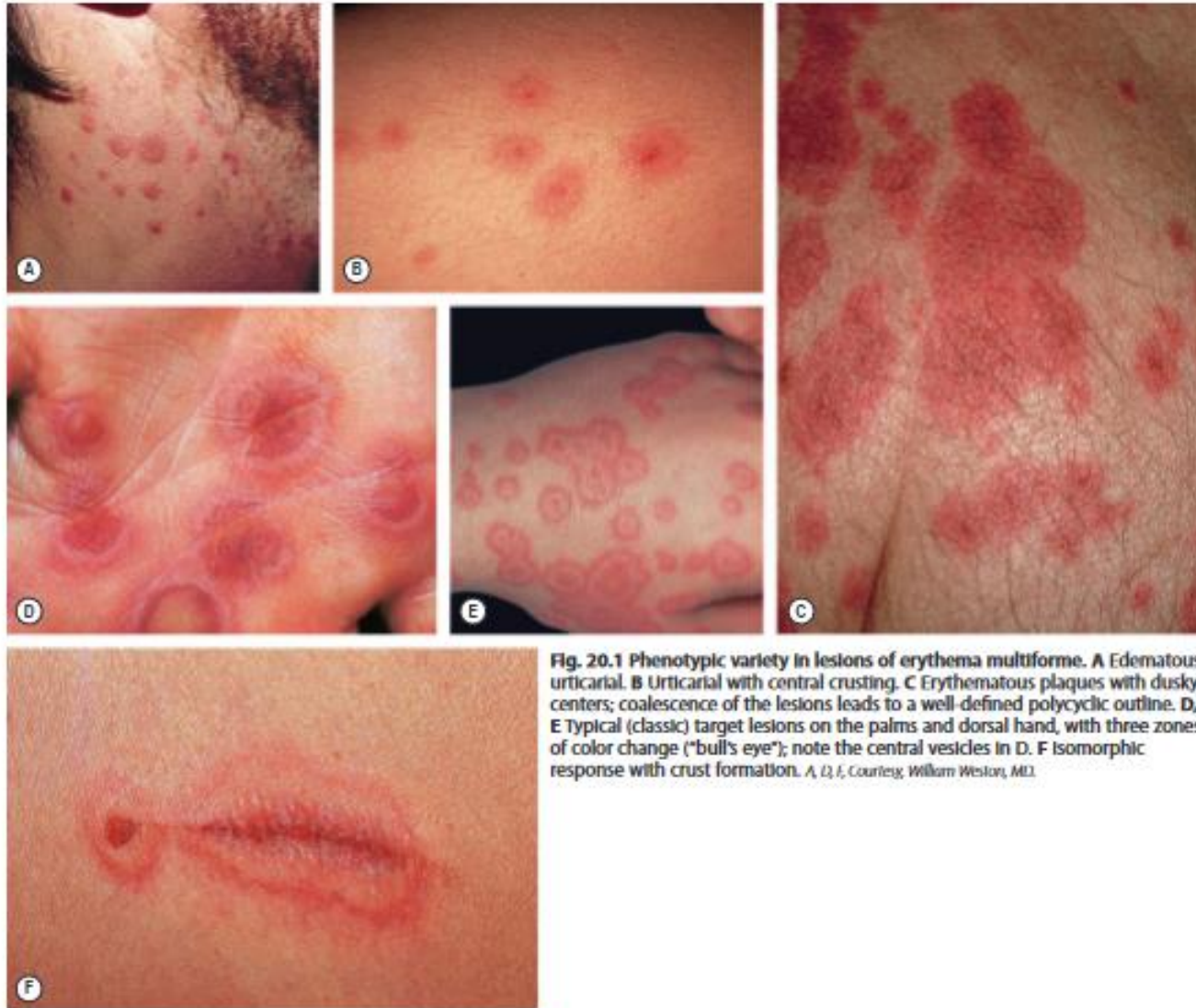
PRECIPITATING FACTORS IN ERYTHEMA MULTIFORME

Infections (approx. 90% of cases)	Viral	• Herpes simplex virus (HSV-1, HSV-2) • Parapoxvirus (orf) • Vaccinia (smallpox vaccine) • Varicella zoster virus (chickenpox) • Adenovirus • Epstein–Barr virus • Cytomegalovirus • Hepatitis virus • Coxsackievirus • Parvovirus B19 • Human immunodeficiency virus
	Bacterial	• <i>Mycoplasma pneumoniae</i>* • <i>Chlamydophila</i> (formerly <i>Chlamydia</i>) <i>psittaci</i> (ornithosis) • <i>Salmonella</i> • <i>Mycobacterium tuberculosis</i>
	Fungal	• <i>Histoplasma capsulatum</i> • Dermatophytes
Drugs (unusual)		Primarily: • Nonsteroidal anti-inflammatory drugs • Sulfonamides • Anticonvulsants • Other antibiotics, e.g. aminopenicillins • Allopurinol
Exposures (unusual)		• Poison ivy
Systemic disease (rare)		• Inflammatory bowel disease • Lupus erythematosus[†] (Rowell syndrome) • Behçet disease[†]



N-B

- **Target lesions:**
- - Typical: have 3 zones and well-defined.
- - Atypical: have only 2 zones and/or poorly-defined. May be:
 - Raised (Papular): palpable, seen in EM.
 - Flat (Macular): non- palpable, seen in SJS / TEN, but not EM.



Skin lesion	Typical target lesion (iris , bulles eyes 3 zones Atypical popular (2 zones)
Skin distribution	upper extremities - especially the dorsal aspects of the hands and the forearms , palms, neck, face and trunk. • Koebner phenomenon
Mucosal affection	absent or mild in EM minor. Severe in EM major. Vesiculo-bullous >>> painful erosions. • Affects oral , ocular and genital mucosae
Systemic affection	Em Minor :no Em major arthralgia , renal , pulmonary

5- clinical variants	
Em minor	
Em major	
Localized vesiculobullous	
Rowelsyndrom	LE + EM like lesion
Atypical cases	1. Persistant EM 2. EM along blaschko line



Histopathology

- in EM , the keratinocyte is the target of the inflammatory insult
- Basket-wave stratum corneum
 - necrotic keratinocytes
 - Spongiosis
 - Vacuolar degeneration of the basal cell layer,
- Papillary dermal edema
- Superficial peri-vascular lymphocytes (may obscure the
- Extravasated erythrocytes

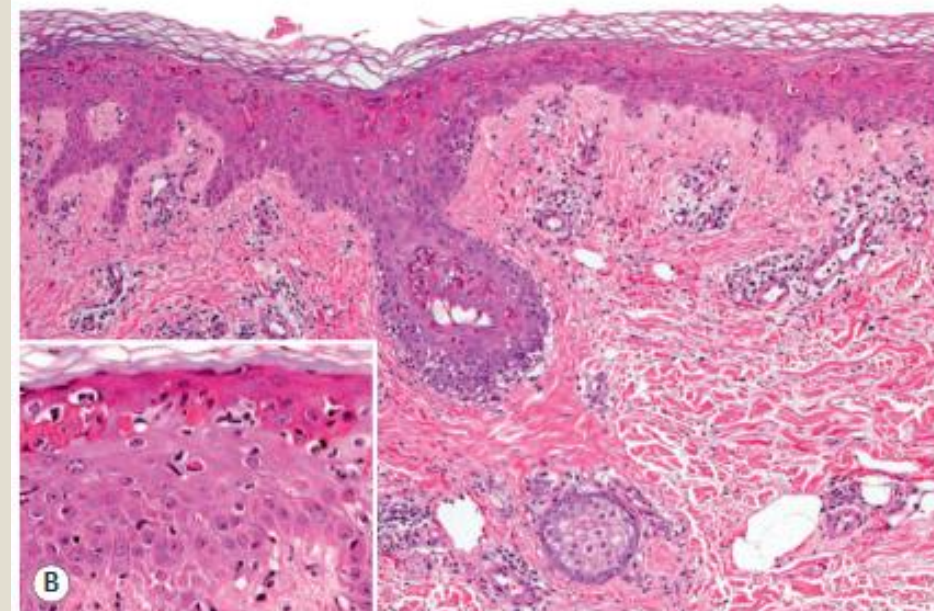
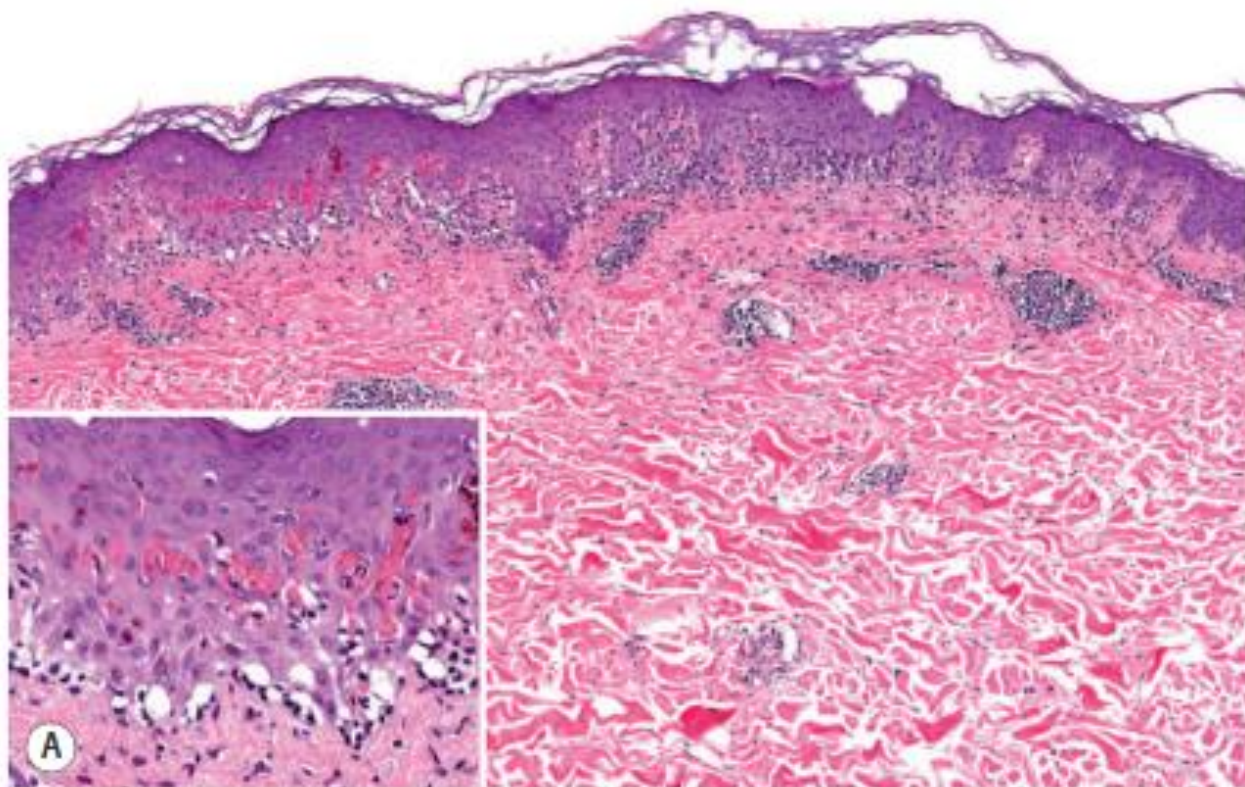


Fig. 20.4 Erythema multiforme – histopathologic features. **A** Early lesion – focal sites of apoptosis of keratinocytes with an interface dermatitis and vacuolar degeneration of the basal layer (inset). A perivascular lymphocytic infiltrate is also present. **B** Late lesion – more confluent necrosis of keratinocytes within the upper layers of the stratum spinosum (inset). The hair follicle still has evidence of an interface dermatitis. *Courtesy, Lorenzo Cerroni, MD.*

DD

- Conditions with target-like” or “targetoid ” lesions;
 - Sweet’s syndrome
 - Acute hemorrhagic edema of infancy
 - Paraneoplastic pemphigus
 - Allergic contact dermatitis
 - Linear immunoglobulin A bullous dermatosis
 - Subacute cutaneous LE
 - Fixed drug eruption
 - Kawasaki disease
 - Vasculitis
 - Erythema annulare centrifugum

COMPARISON OF URTICARIA AND ERYTHEMA MULTIFORME

Urticaria	Erythema multiforme
Central zone is normal skin or transient duskiness	Central zone is damaged skin (dusky, bullous or crusted)
Individual lesions are transient, lasting <24 hours*	Individual lesions "fixed" for at least 7 days
New lesions appear daily	All lesions appear within first 72 hours
May be associated with swelling of face, hands or feet (angioedema)	No edema
*Can be documented by circling individual lesions with ink then observing over time.	

COMPARISON OF ERYTHEMA MULTIFORME (EM) MINOR, EM MAJOR, STEVENS–JOHNSON SYNDROME (SJS), SJS/TOXIC EPIDERMAL NECROLYSIS (TEN) OVERLAP, AND TEN						
	Type of skin lesions	Distribution	Mucosal involvement	Systemic symptoms	Progression to TEN	Precipitating factors
EM minor (Figs 20.1 & 20.2)	• Typical targets • $\pm$ Papular atypical targets	Extremities (especially elbows, knees, wrists, hands), face	Absent or mild	Absent	No	• Herpes simplex virus • Other infectious agents
EM major (Fig. 20.3)	• Typical targets • $\pm$ Papular atypical targets • Occasionally bullous lesions	Extremities, face	Severe	Usually present • Fever • Arthralgias	No	• Herpes simplex virus • <i>Mycoplasma pneumoniae</i> • Other infectious agents • Rarely, drugs
SJS (Figs 20.7 & 20.14)	• Dusky and/or dusky-red macules with epidermal detachment and erosions • Macular atypical targets • Bullous lesions • <10% BSA detachment	Isolated lesions Confluence (+) Trunk, face	Severe	Usually present • Fever • Lymphadenopathy • Hepatitis • Cytopenias	Possible	• Drugs • Occasionally, <i>Mycoplasma pneumoniae</i>* • Rarely, immunizations
SJS/TEN overlap (Fig. 20.11)	• Similar to SJS • 10–30% BSA detachment	Isolated lesions Confluence (++) Trunk, face, neck	Severe	• Same as SJS	Likely	• Drugs
TEN (Figs 20.8 & 20.9)	• Similar to SJS • >30% BSA detachment	Confluence (+++) Isolated lesions rare Trunk, face, neck, elsewhere	Severe, with involvement of respiratory and gastrointestinal mucosa	• Same as SJS plus nephritis		• Drugs
*Occasionally, also subcorneal pustules.						

Treatment

- Treatment of the acute eruption:
 - Topical therapy
 - Good nursing is very important.
 - Topical antiseptics
- Systemic therapy
 - Oral antihistamines for 3 - 4 days
 - Systemic corticosteroids
 - A Used for or more severe cases
 - A Prednisone 0.5-1 mg/kg/ d

Recurrent disease:

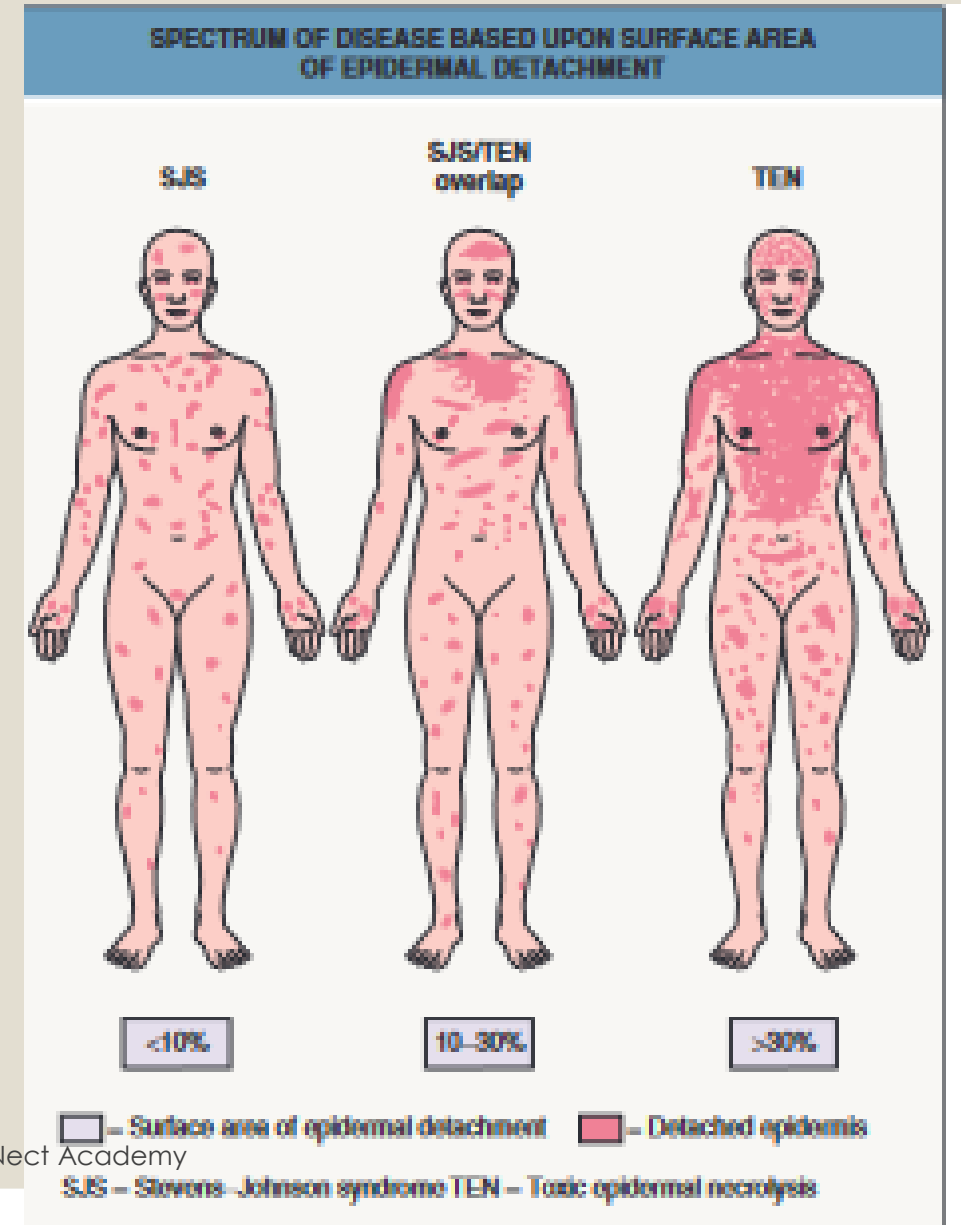
prophylaxis for at least 6 months with
oral acyclovir 200 mg three times daily),

Stevens-Johnson syndrome (SJS) toxic epidermal necrolysis (TEN)

- **acute severe life-threatening muco-cutaneous cytotoxic reactions to drug .**
- extensive keratinocyte cell death that results in the separation at the DEJ

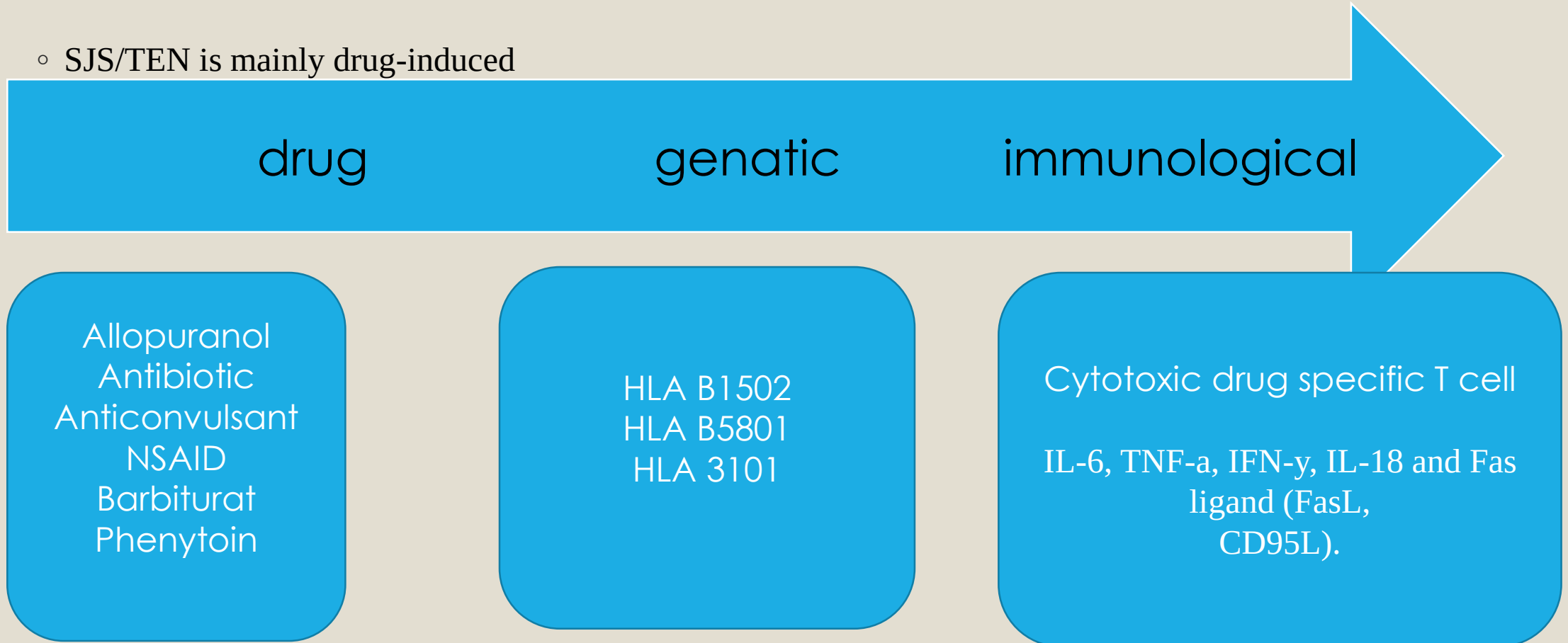


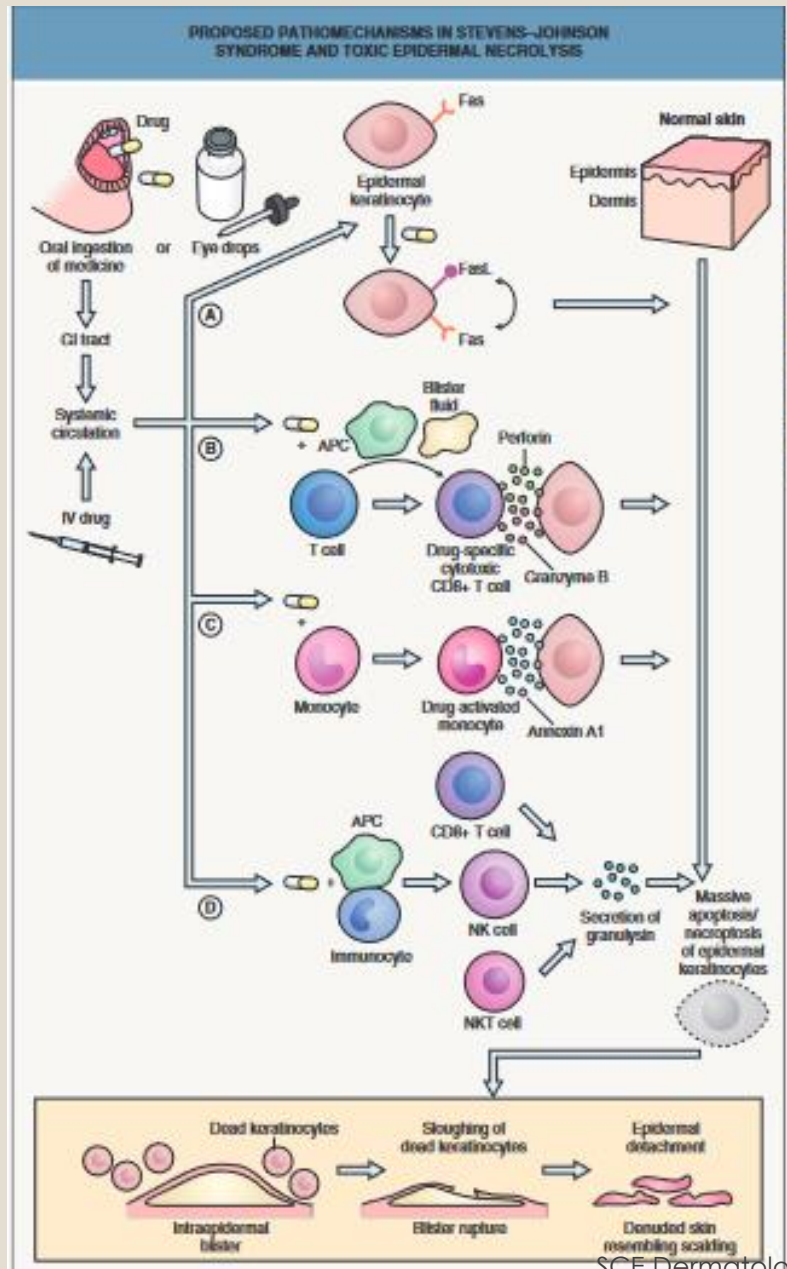
- SJS is the less extensive form and TEN is the more extensive form.
- SJS : 10%
- TEN :30%
- SJS – TEN :10-30%



aetiopathogenesis

- SJS/TEN is mainly drug-induced





- impaired capacity to detoxify reactive drug metabolites formation of antigenic complex (drug metabolites + host tissues) immune response.
- may be triggered by infections, most notably by *Mycoplasma pneumoniae*.
- **Pathogenesis of SJS/TEN (in brief):**
 - Exposure to certain drugs in an individual with genetic susceptibility ■=>specific immune reaction to the drug or one of its metabolites ■=>strong expression of the cytolytic molecule
 - Tips- (FasL, perforin / granzyme B, granuleysin) C> apoptosis of keratinocytes and subsequent epidermal necrosis and detachment.

STEVENS-JOHNSON SYNDROME (SJS) AND TOXIC EPIDERMAL NECROLYSIS (TEN): EPIDEMIOLOGY AND RISK FACTORS

Annual incidence	1.2–6 per million (SJS)
	0.4–1.2 per million (TEN)
Ratio women:men	1.5:1
Risk factors	Slow acetylator genotypes
	Immunosuppression (e.g. HIV infection, lymphoma)
	Concomitant administration of radiotherapy and anticonvulsants (most commonly, those with brain tumors)
	HLA-B*15:02 – Asians (in particular Han Chinese, Thai, Malaysian populations) [†] and East Indians exposed to carbamazepine
	HLA-B*15:02 – Han Chinese/lamotrigine
	HLA-B*15:02 – Han Chinese/phenytoin
	HLA-A*31:01 – Europeans/carbamazepine
	HLA-B*58:01 – Han Chinese/allopurinol
[†] The background prevalence of HLA-B*15:02 varies from 4% to 15% in affected populations but is extremely rare in Northern Europeans (<0.01%).	

MEDICATIONS MOST FREQUENTLY ASSOCIATED WITH STEVENS-JOHNSON SYNDROME AND TOXIC EPIDERMAL NECROLYSIS

Allopurinol
Aminopenicillins
Amithiozone (thioacetazone)*,†
Antiretroviral drugs, especially NNRTIs (e.g. nevirapine, efavirenz, etravirine)
Barbiturates
Carbamazepine
Cephalosporins
Checkpoint inhibitors (e.g. ipilimumab, nivolumab)
Chlormezanone*,‡
Phenytoin anticonvulsants
Lamotrigine (risk may be increased when co-administered with valproic acid)
Phenylbutazone*,§
Piroxicam
Quinolones
Sulfonamide antibiotics, including sulfamethoxazole**, sulfadiazine*,†
Sulfasalazine

*Not available in the US.

†Antibacterial.

‡Sedative/hypnotic.

§Nonsteroidal anti-inflammatory drug.

**Usually in combination with trimethoprim.

Clinical picture

- upper trunk and face
- The earliest skin lesions are **tender atypical target lesions** (erythematous, dusky red or purpuric macules) —> increase in size and number —> coalesce —> large areas of confluent erythema.
- **The dusky red color changes into characteristic gray hue** (= full thickness necrosis of the epidermis).
- **the necrotic epidermis detaches from the underlying dermis** —
 - *■ vesicles (necrolytic blistering).>>>>denuded dermis (may bleed and /secondarily infected).



Fig. 20.8 Cutaneous features of toxic epidermal necrolysis (TEN). Characteristic dusky red color of the early macular eruption in TEN. Lesions with this color often progress to full-blown necrolytic lesions with dermal-epidermal detachment.

- upper trunk and face
- The distal portions of the arms and the legs are relatively spared
- palms and soles with atypical target lesions
- Nikolsky sign is positive
- positive Asboe-Hansen sign



Fig. 20.11 Stevens–Johnson syndrome (SJS)/toxic epidermal necrolysis (TEN) overlap and early TEN. **A** In addition to mucosal involvement and numerous dusky lesions with flaccid bullae, there are areas of coalescence and multiple sites of epidermal detachment. Because the latter involved >10% body surface area, the patient was classified as having SJS/TEN overlap. **B** Close-up of epidermal detachment, whose appearance has been likened to wet cigarette paper. **C** A child with early TEN due to diclofenac. There is coalescence of the

- Mucosal lesions:
 - Erythema and erosions of the buccal, ocular and genital mucosae are
 - Respiratory tract: cough, chest pain, dyspnoea or haemoptysis
 - Gastrointestinal tract: diarrhea
- Systemic manifestations:
 - hepatitis and cytopenias



Fig. 20.7 Mucosal involvement in Stevens–Johnson syndrome. **A** Erythema and conjunctival erosions. **B** Erosions of the genital mucosa.

SCORTEN	
Prognostic factors	Points
Age >40 years	1
Heart rate >120 bpm	1
Cancer or hematologic malignancy	1
BSA involved on day 1 >10%	1
Serum urea level (>10 mmol/l)	1
Serum bicarbonate level (<20 mmol/l)	1
Serum glucose level (>14 mmol/l)	1
SCORTEN	Mortality rate (%)
0-1	3.2
2	12.1
3	35.8
4	58.3
≥5	90



Fig. 20.9 Clinical features of toxic epidermal necrolysis (TEN). **A** Detachment of large sheets of necrolytic epidermis (>30% body surface area), leading to extensive areas of denuded skin. A few intact bullae are still present. **B** Extensive symmetric hemorrhagic crusting of the face with areas of denudation. **C** Epidermal detachment of palmar skin. **D** Note the rolled and folded sheets of detached epidermis at the edge of denuded skin in addition to widespread erythema and intact bullae. *D, Courtesy, Luis Requena, MD.*

Complication

Acute

Hypothermia
Fluid loss
Anaemia
Hypoalbuminamia
Hyperglycemia
Septicemia
Hepatic failure

Chronic

Skin :

Scaring ,
dyspigmentation
Nail shedding
TE , photosensitivity

Ocular :

Ulcer , scar ,
ectrobion

GIT :

Oesophogeal
stricture

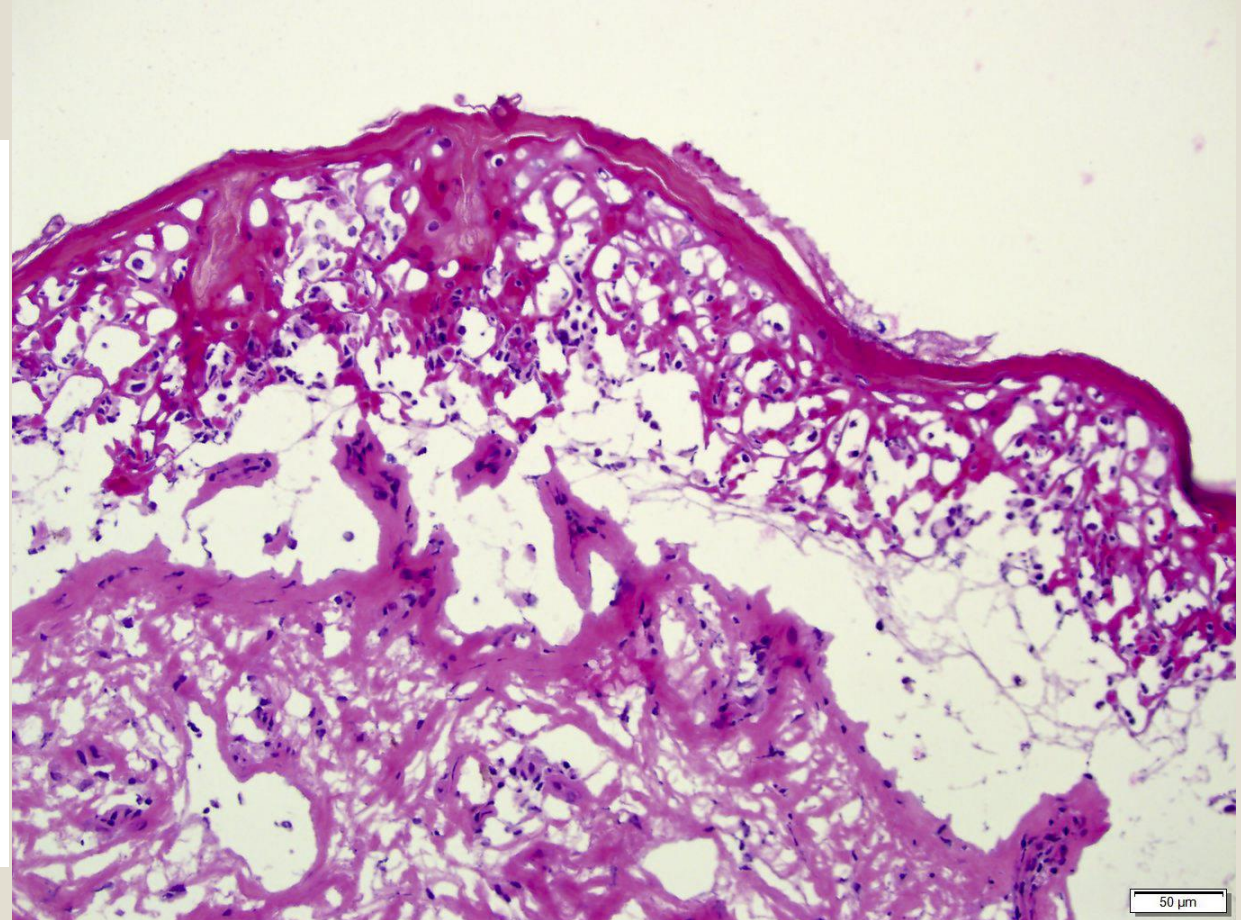
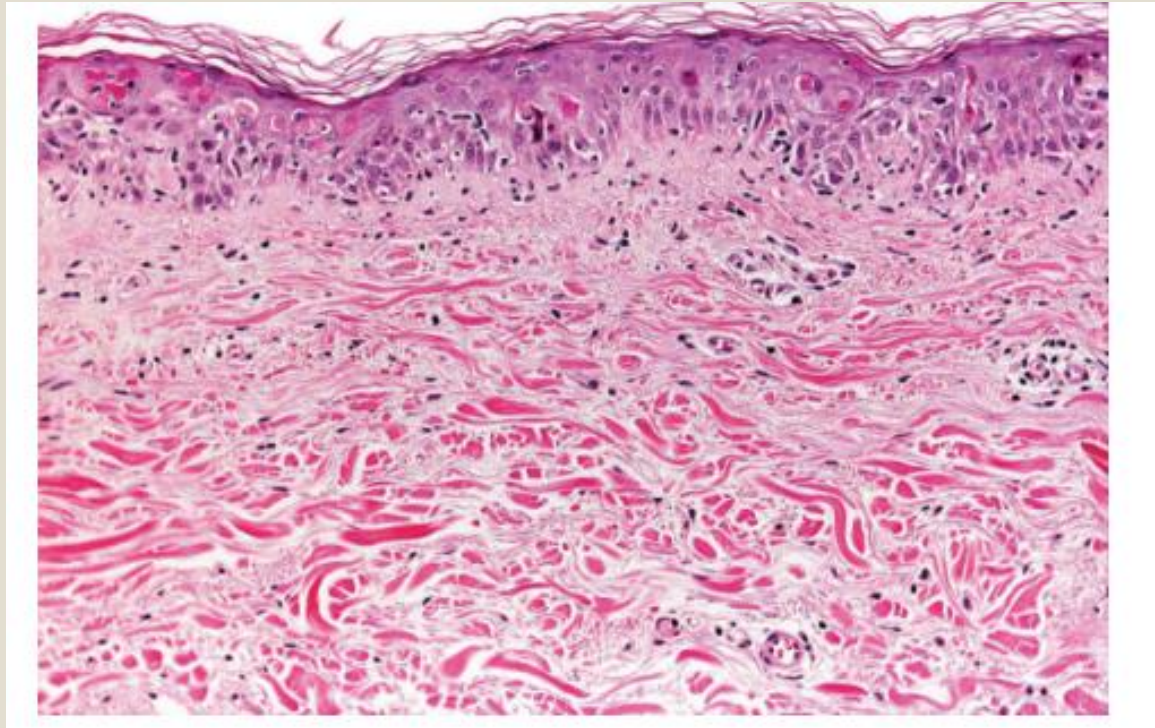


Fig. 20.12 Sequelae of toxic epidermal necrolysis. A Symblepharon, erosion of lower lateral eyelid margin and sparse eyelashes; the patient also had entropion with an ingrowth of eyelashes and pebble-like scarring of facial skin. B Larger irregular areas of hypopigmented scarring. C Nail dystrophy consisting of longitudinal ridging and fissuring, fragility, and distal notching.

Histopathology

○

- Early lesions of SJS/TEN: apoptotic keratinocytes scattered in the basal and supra-basal layers of the epidermis.
- At later stages: confluent necrosis of the entire epidermis, sub-epidermal blister, and a sparse perivascular lymphocytic infiltrate (CD 8+ by immunohistochemistry).



DD

DIFFERENTIAL DIAGNOSIS OF TOXIC EPIDERMAL NECROLYSIS	
Disease	Clinical and histologic features
Erythema multiforme and Stevens-Johnson syndrome	• See Table 20.1
Staphylococcal scalded skin syndrome (SSSS; see Chs 74 & 81)	• Children and occasionally adults • Split is more superficial so base is eroded epidermis not dermis • Perioral and periorbital accentuation with radial scale-crusts; no intraoral lesions • <i>Microscopically, subcorneal split with a normal underlying epidermis</i>
Invasive fungal dermatitis (see Ch. 34)	• Very-low-birth-weight newborns • Widespread erythema with erosions resembling a thermal burn • Usually due to candidal infection
Acute generalized exanthematous pustulosis (AGEP; see Ch. 21)	• Numerous small, non-follicular pustules; can coalesce leading to large areas of exfoliation • Peripheral neutrophilia • <i>Microscopically, subcorneal pustules; no full-thickness epidermal necrosis</i>
Generalized fixed drug eruption (see Ch. 21)	• When there are numerous mucocutaneous lesions • Timing post drug exposure differs • <i>Microscopically, significant overlap</i>
Drug-induced linear IgA bullous dermatosis (LABD), paraneoplastic pemphigus (PNP), bullous pemphigoid (see Chs 29–31)	• Exposure to high-risk drugs, e.g. vancomycin for LABD • <i>Findings by routine histology and especially direct and indirect (PNP) immunofluorescence microscopy allow distinction</i>
Lupus erythematosus (Rowell syndrome; see Ch. 41)	• Usually resembles EM but can have TEN-like presentation • Appears <i>de novo</i> or as therapy tapered • <i>Microscopically, significant overlap</i>
Severe acute GVHD (stage/grade IV; see Ch. 52)	• Recipient of allogeneic hematopoietic stem cell transplant • Uncommon due to use of prophylactic GVHD therapies • <i>Microscopically, significant overlap</i>
Drug reaction with eosinophilia and systemic symptoms (DRESS; see Ch. 21)	• In addition to widespread erythema, vesiculobullae and pustules can develop • Facial edema; peripheral eosinophilia (most patients)

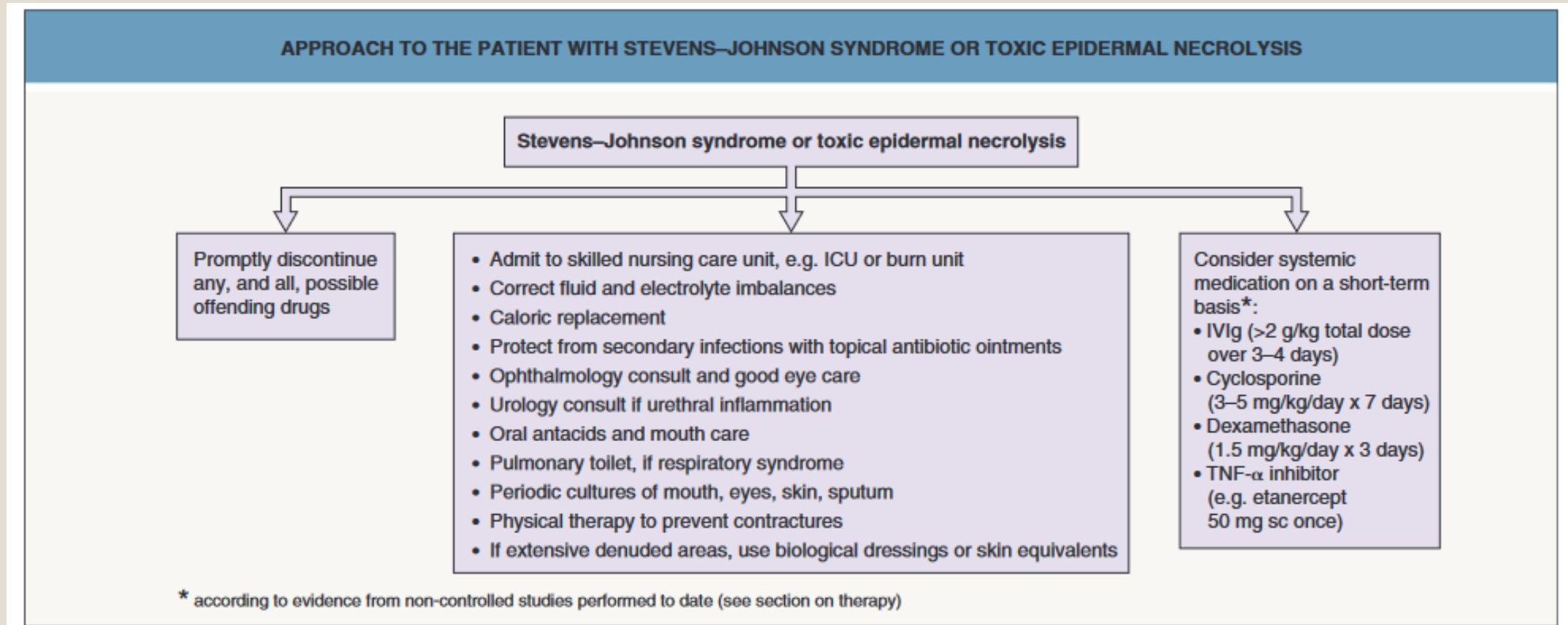
○ Other investigation

Swabs from lesional skin

- Chest X -ray (mandatory).
- Blood tests: CBC, ESR, CRP, urea and electrolytes,



treatment



Erythemas

- Erythema is a change in the color of the skin due to dilation of blood vessels especially those in papillary and reticular dermis

- **Types**

Generally, there are four main types of erythema:

1. Diffuse e.g. scarlet fever.
2. Regional: .
3. Reticulated e.g. cutaneous polyarteritis nodosa.
4. Figurate:

Figurate erythemas



Erythema annular
centrifugum



Erythema
marginatum



Erythema
chronicum migrans



Erythema gyratum
repens

Erythema annulare centrifugum (E A C)

- any age (peak incidence in 5th decade / middle age)
 - equal sex
- Fungi: dermatophytes (particularly, tinea pedis), Candida, Penicillium (in blue cheese)
 - Viruses: poxvirus, EBV, Epstein-Barr vims, HIV, herpes simplex,
 - Bacterial: Mycobacteria, streptococci, Escherichia c o li, syphilis
 - 2. Drugs: e.g. diuretics, NSAIDs, antimalarials, gold
 - 3. Neoplasms: e.g. lymphomas, l
 - 4. Foods: blue cheese
- Types
 1. Superficial EAC
 2. Deep EAC

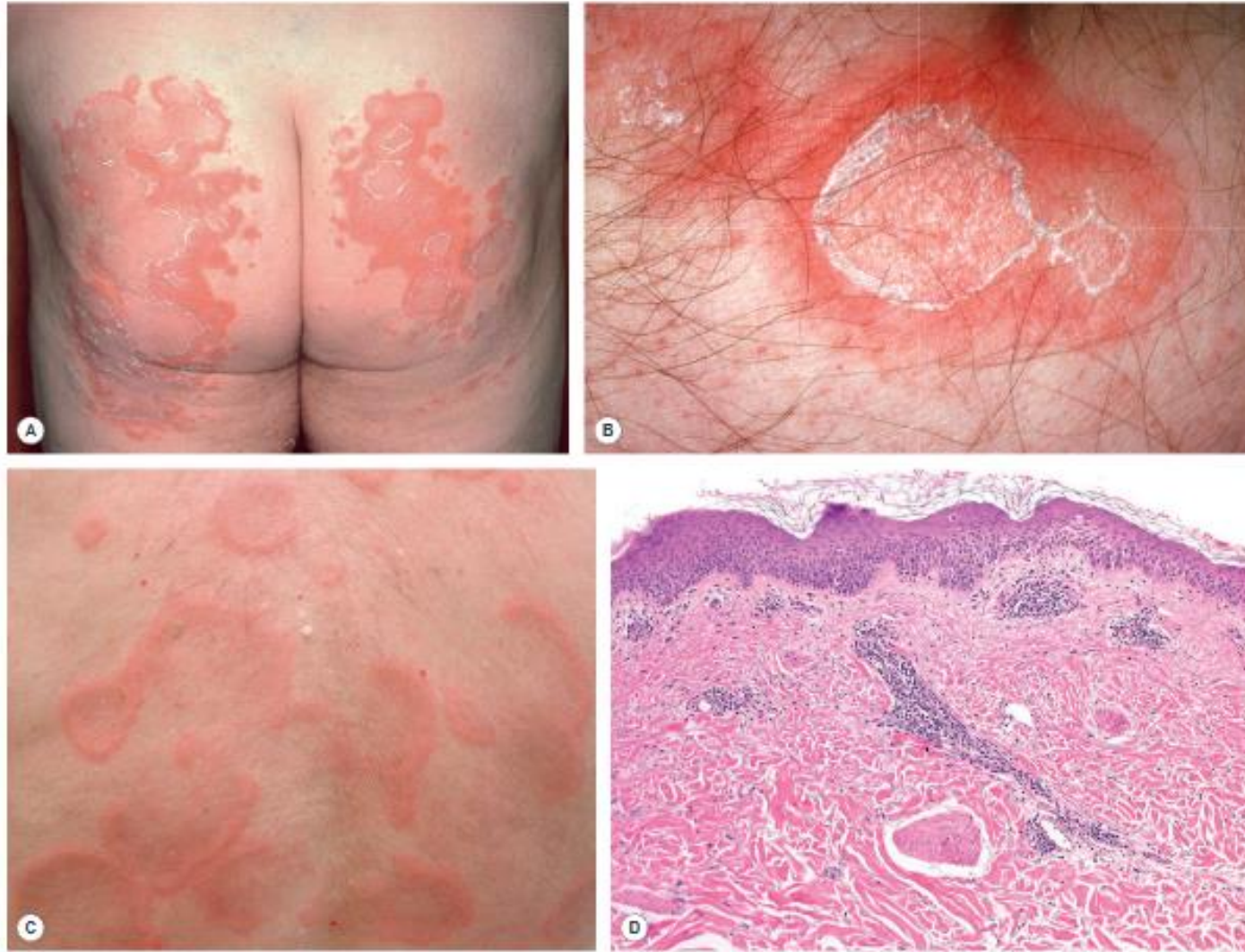


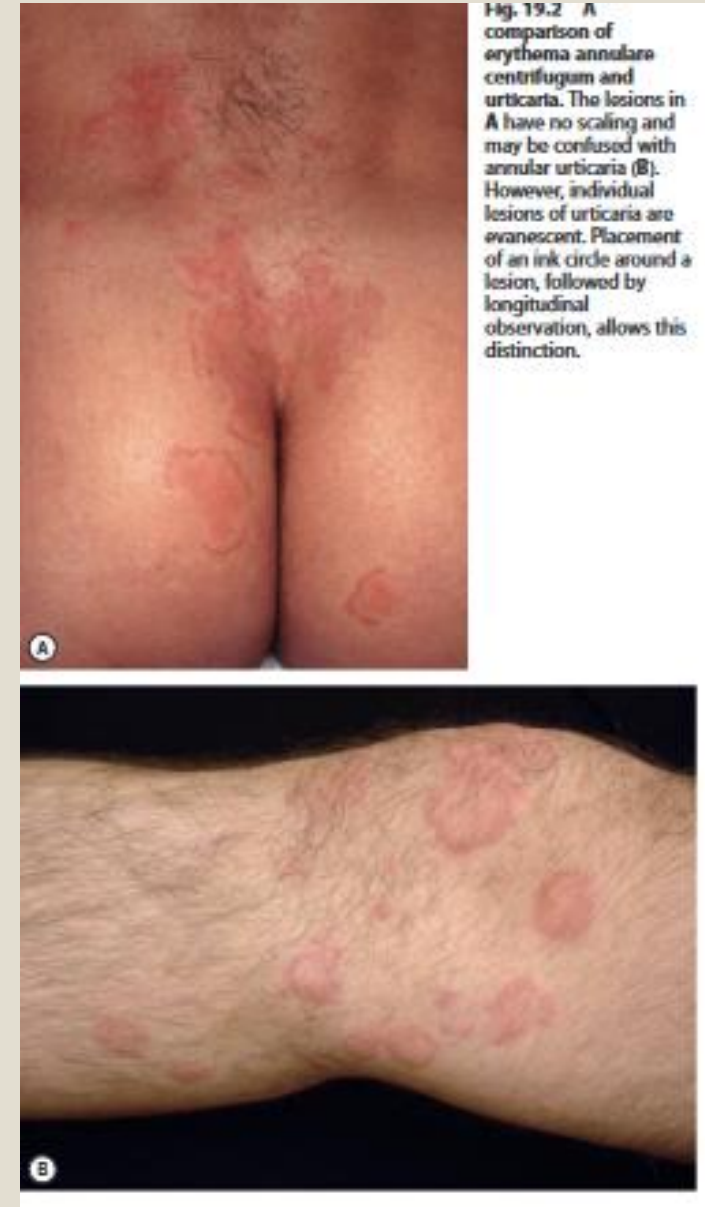
Fig. 19.1 Erythema annulare centrifugum. **A,B** Superficial form, with polycyclic plaque and delicate scale on the inner margin of the advancing edge (trailing scale). The scale is detached centrally. **C** Deep form, with obvious elevation of advancing edges and without trailing scale. **D** Photomicrograph of the superficial form with dermal perivascular lymphohistiocytic infiltrates; note the tight "coat sleeve"-like appearance of the infiltrate. *D, Courtesy, Lorenzo Cerroni, MD.*

	Suprficial EAC	Deep EAC
	Pruritis	No pruritis
	annular erythematous plaque. • incomplete arcs, polycyclic or festooned buttocks, thighs, lower and upper limbs	
	White scaling at the inner margin. <trailing <ul="" scale=""> • Lesions are minimally elevated </trailing>	No scales • Elevated, infiltrated and indurated advancing edge (cord-like)
	NO associated systemic manifestations	

H,P	epidermis: Focal parakeratosis (scale) Mild spongiosis (pruritus)	Epidermis: NORMAL (no scale, no pruritus)
HP DERMIS	superficial perivascular lympho histiocytic infiltrate forming a tight aggregate around vessels Coat sleeve appearance	The cell infiltrate favors the mid and lower dermis
	Tinea corporis (circinata) Annular psoriasis	Annular urticaria

treatment

- Tt of causes
- Sedating antihistaminic
- Topical steroid
- Systemic steroid
- Phototherapy



Eruthema marginatum

- cutaneous manifestation of acute rheumatic fever, one of the major criteria.
- children , equal sex
- Developing countries
- antigenic mimicry (epitopes cross-reacting with group A streptococcal antigens)



- Asymptomatic erythematous macules / patches with rapid peripheral spread
- >>>> serpiginous, poly cyclic or annular plaques
- Transient , evanescent
- Trunk axilla
- No specific HP
- Causes of death :
 - Carditis , renal failure
- Ttt of cause



Fig. 19.3 Erythema marginatum. Polycyclic and evanescent annular lesions are seen on the trunk of this young

Erythema chronicum migrans

- Initial manifestation of lyme disease
- At site of bite by borrelia infected ticks
- Ixodes ticks
- Any age ,equal sex
- Usa , Europe



STAGES AND MAJOR ORGAN MANIFESTATIONS OF LYME DISEASE			
Organ	Early localized disease	Early disseminated disease	Chronic disease
Skin	Erythema migrans		Acrodermatitis chronica atrophicans (Europe)
	Disseminated erythema migrans Borrelial lymphocytoma (Europe)		
Nervous system		Lymphocytic meningitis Cranial neuritis Bell palsy Radiculoneuropathy	Encephalopathy Encephalomyelitis Neuropathy
Musculoskeletal system		Arthralgias, arthritis Myositis	Chronic arthritis
Heart		Atrioventricular block Myopericarditis, pancarditis	
Lymphatic	Regional lymphadenopathy	Regional or generalized lymphadenopathy	
Other		Conjunctivitis, iritis Hepatitis Non-productive cough Microscopic hematuria or proteinuria	

- Spirochetal lipoproteins >>> trigger immune response
- c/p :
 - Erythematous expanding annular plaque
 - Bull eyes (targetoid lesion)
 - Spirotrichosis pattern
 - Trunk , axilla , groin , popliteal fossa
 - No specific HP
 - Investigation for borrelia

DD: FDE , cellulitis

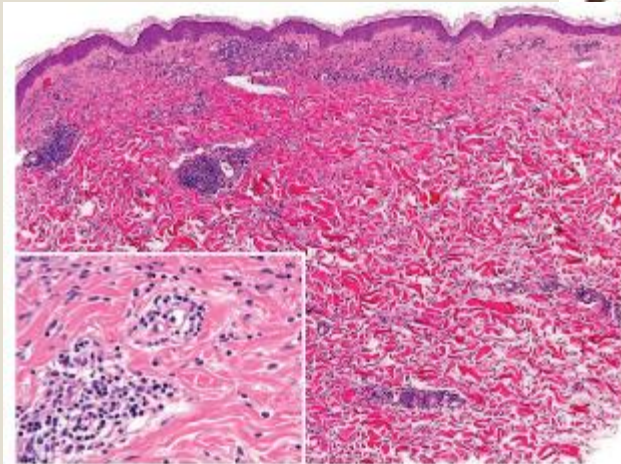


FIGURE 3 Erythema migrans (EM) with a dense perivascular infiltrate is present within the dermis as well as a scattered interstitial pattern. The infiltrate is composed of lymphocytes and plasma cells (insert). Courtesy, Lorenzo Cerioni, MD.

2011 CDC LYME DISEASE SURVEILLANCE CASE DEFINITION

Definition of erythema migrans (EM)

A skin lesion that typically begins as a red macule or papule and expands over a period of days to weeks to form a large round lesion, often with partial central clearing. A single primary lesion must reach ≥ 5 cm in diameter. Secondary lesions also may occur. Annular erythematous lesions occurring within several hours of a tick bite represent hypersensitivity reactions and do not qualify as EM. The diagnosis of EM must be made by a physician.

Confirmed

- A case of EM with a known exposure*
- A case of EM with laboratory evidence of infection[†] and without a known exposure
- A case with at least one late manifestation[‡] that has laboratory evidence of infection

Probable

- Any other case of physician-diagnosed Lyme disease that has laboratory evidence of infection

Suspected

- A case of EM where there is no known exposure* and no laboratory evidence of infection
- A case with laboratory evidence of infection but no clinical information available (e.g. a laboratory report)

TREATMENT OPTIONS FOR LYME DISEASE				
Manifestation	Drug	Dose (pediatric dose*)	Duration (days)	Comments
Erythema migrans	Doxycycline	100 mg (2 mg/kg) po BID	14 (range, 10–21)	• Do not use in children <8 yrs of age, or pregnant or lactating women • Good CNS penetration • Also treats human anaplasmosis
	Amoxicillin	500 mg (16 mg/kg) po TID	14 (range, 14–21)	
	Cefuroxime axetil	500 mg (15 mg/kg) po BID	14 (range, 14–21)	
Manifestation	Drugs		Duration (days)	Comments
Meningitis	Ceftriaxone (iv), cefotaxime (iv)		14 (range, 10–28)	
Cranial nerve palsy (without meningitis)	Same drugs and dosages as for erythema migrans		14 (range, 14–21)	• May not affect outcome of facial palsy
Carditis	If symptomatic, initiate treatment with parenteral antibiotic (e.g. ceftriaxone) then based upon clinical response, finish course with oral agents		14 (range, 14–21)	• Advanced heart block may require a pacemaker
Arthritis	Same parenteral agents as for meningitis and same oral agents as for erythema migrans		28	• If persists or recurs, a second course of antibiotics

*The maximum pediatric dose is the adult dose.

Erythema gyratum reprens

- Paraneoplastic
- Adult male >40 years
- Carcinoma of lung
- Immune cross reaction between tumor antigen and cutaneous antige



Fig. 19.8 Erythema gyratum repens. Multiple gyrate erythematous plaques, showing a wood-grain pattern. Courtesy, Agustín Alomar, MD.

- C/P:
 - Multiple annular erythematous lesion >>>
 - Rapid progressive 1 cm /day
 - Concentric , wood grain , zebra like
 - Severe pruritis
- Additional feature
 - Acquired ichthyosis
 - Ppk



- No specific HP
- DIF: granular IgG, C3

Investigation detect neoplasm

DD:

Tinea imprecata, EM

- TTT:
 - RESOLVE WITH TT OF NEOPLASM



DIFFERENTIAL DIAGNOSIS OF FIGURATE ERYTHEMA			
Disorders	Clinical features	Histologic findings	Cross references
<i>Transient individual lesions (typically lasting <24 hours)</i>			
Urticaria	Transient and recurrent wheals; pruritus	Superficial perivascular infiltrate of eosinophils and lymphocytes; occasionally neutrophils	Ch. 18
Urticaria multiforme (a variant of annular urticaria)	Annular and polycyclic wheals, often with central ecchymosis that should not be confused with necrosis; pruritus; edema of the face, hands, feet; absence of mucous membrane erosions or crusts; recent viral or bacterial infection	Indistinguishable from other forms of urticaria	
Serum sickness-like reaction	Annular and polycyclic wheals often have a purpuric component; acral edema; high fevers, arthralgias, lymphadenopathy; drug exposure (e.g. cefaclor, penicillins, minocycline)	Similar to urticaria; unlike serum sickness (e.g. due to equine- or rabbit-derived antithymocyte globulin), leukocytoclastic vasculitis is usually absent	
Erythema marginatum	Polycyclic erythematous eruption; antecedent group A streptococcal infection; additional manifestations of acute rheumatic fever, e.g. arthritis and carditis	Superficial perivascular infiltrate of neutrophils and lymphocytes; occasionally eosinophils	See text
Erythema marginatum-like lesions of hereditary angioedema	Transient polycyclic erythematous eruption; favors trunk; precedes or accompanies episodes of angioedema	Minimal to no inflammatory infiltrate	Ch. 18
African trypanosomiasis	Annular erythematous patches on the trunk associated with fever spikes	Perivascular lymphocytic infiltrate with many plasma cells; parasites are occasionally seen within the dermis	Ch. 83
Erythrokeratoderma varians	Transient gyrate or circinate erythematous patches (usually fade/migrate within a few hours); also fixed figurate hyperkeratotic plaques; GJB4 mutations	Scant perivascular lymphocytic infiltrate; dilated capillaries; psoriasiform acanthosis and mild hyperkeratosis (plaques)	Ch. 57
<i>Urticarial lesions lasting >24 hours</i>			
Allergic urticarial eruption	Annular edematous plaques; favors trunk	Edema; superficial and deep lymphocytic infiltrate with eosinophils	
Annular erythema of infancy (including neutrophilic figurate erythema of infancy)	Recurrent annular erythematous plaques, usually lasting a few days; typically affects infants 3–11 months of age (see Fig. 19.4)	Perivascular lymphocytes and eosinophils; sometimes a predominantly neutrophilic infiltrate	
Wells syndrome	Edematous erythematous plaques, often annular or arcuate with a greenish color centrally; pruritus; peripheral blood eosinophilia	Diffuse infiltrate of eosinophils; flame figures	Ch. 25
Urticarial vasculitis	Urticarial plaques, often annular, that may have a purpuric or petechial component; pain, burning and/or pruritus; systemic manifestations in hypocomplementemic form	Leukocytoclastic vasculitis	Ch. 24
Erythema migrans	Erythema slowly migrates from site of tick bite; influenza-like syndrome; secondary distant lesions	Superficial and deep perivascular lymphocytic infiltrate; eosinophils and plasma cells in erythema migrans; mucin in LE tumidus and annular erythema of Sjögren syndrome	See text and Ch. 74
Erythema annulare centrifugum (deep)	Annular erythema with raised borders		See text
Lymphocytic infiltrate of Jessner	Annular erythematous plaques; favors face and upper trunk		Ch. 121
LE tumidus	Annular erythematous plaques; favors face, upper trunk > extensor forearms		Ch. 41
Annular erythema of Sjögren syndrome*	Annular erythematous plaques; favors face, arms, upper trunk; most common in Asian patients; anti-Ro antibodies		Ch. 45
<i>Granulomatous lesions</i>			
Granuloma annulare	Skin-colored to pink annular and arcuate plaques with well-circumscribed central papules; favors acral and extensor sites	Palsading granulomas with altered collagen and fibrin	Ch. 93

DIFFERENTIAL DIAGNOSIS OF FIGURATE ERYTHEMA			
Disorders	Clinical features	Histologic findings	Cross references
Granulomatous lesions			
Annular elastolytic giant cell granuloma	Pink annular plaques with atrophy and hypopigmentation centrally; favors sites of chronic sun exposure	Granulomatous infiltrate with elastophagocytosis by giant cells (border of lesion); absence of elastic fibers (center of lesion)	Ch. 93
Interstitial granulomatous dermatitis (IGD); interstitial granulomatous drug reaction (IGDR)	Annular erythematous plaques (ex. in IGD, linear cords); favors axilla, groin, lateral trunk and legs; associated arthralgia/arthritis; IGDR is most often due to antihypertensive and lipid-lowering agents	Pale-staining granuloma surrounding tiny foci of fragmented collagen; neutrophils and sometimes eosinophils; little to no mucus	Ch. 45
Sarcoidosis	Red-brown annular plaques and papules; favors face; systemic manifestations	Sarcoid granulomas ("naked tubercles")	Ch. 93
Borderline or tuberculoid leprosy	Annular plaques with raised erythematous borders and (in tuberculoid lesions) central hypopigmentation; asymmetric distribution; hyposthesia, alopecia	Linear or oval perineural granulomas in the dermis; foamy macrophages and AFB in borderline lesions	Ch. 75
Populosequestrous lesions			
Erythema annulare centrifugum (superficial)	Annular lesions with trailing scale; favors thighs, hips and trunk	Superficial perivascular lymphocytosis with focal spongiosis and parakeratosis; focal hemorrhage and exocytosis of lymphocytes in pityriasis rosea	See text
Erythema gyratum repens	Concentric erythematous rings with wood-grain appearance; erythema moves ~1 cm per day; associated with cancer		See text and Ch. 53
Pityriasis rosea	Herald patch on trunk; oval annular papules and plaques with central fine scaling and/or a collarette of trailing scale; Christmas tree distribution on posterior trunk		Ch. 9
Pityriasis rubra pilaris	Appears as the primary "classic" lesions resolve; resembles erythema gyratum repens	Epithelial hyperplasia; alternating vertical and horizontal ortho- and parakeratosis; follicular plugging with shoulders of parakeratosis	See text and Ch. 9
Tinea corporis	Annular plaques with scale (exception: tinea incognita); pustules in border; concentric lesions in tinea imbricata; KOH positive	Dermatophytes in the stratum corneum	Ch. 77
Lichen planus (LP)	Violaceous annular plaques; gray-brown to brown hypopigmentation centrally; mucosal involvement; photodistribution in actinic LP	Hypogranulosis, band-like lymphocytic infiltrate at dermal-epidermal junction, necrotic keratinocytes	Ch. 11
Annular LE (subacute > discoid), neonatal LE, mothers of boys with chronic granulomatous disease	Photosensitivity; favors face (discoid), anterior upper extremities/upper trunk (SCLE), periorbital area (neonatal LE); anti-Ro antibodies (SCLE, neonatal LE); erythema gyratum atrophicans tensionis neonatale is considered a variant of neonatal LE	Vascular interface dermatitis; often perivascular lymphohistiocytic inflammation	Ch. 41
Seborrheic dermatitis	Erythematous annular plaques with scaling; favors face and central chest	Psoriasisiform epithelial hyperplasia, spongiosis, perivascular lymphocytes	Ch. 13
Psoriasis	Annular scaly plaques with slow expansion	Psoriasisiform epithelial hyperplasia, parakeratosis, diminished granular layer, elongated rete ridges, neutrophils migrating into epidermis	Ch. 8
Ichthyosis linearis circumflexa of Netherton syndrome	Serpiginous or circinate erythematous plaques bordered by double-edged scale; trichothemia innominata; atopic diathesis, elevated serum IgE	Psoriasisiform features with hyperkeratosis and a well-developed granular layer	Ch. 57
Lesions with absence or variable presence of scale			
Annular lichenoid dermatitis of youth	Red-brown annular patches or thin plaques with central hypopigmentation; favors groin and flanks; affects children and young adults	Lichenoid infiltrate; marked keratinocyte necrosis limited to the tips of rete ridges	
Erythema papulatum centrifugum	Usually a single, large annulus on the trunk whose border is 2-6 cm in width and composed of small erythematous papules, papulovesicles or crusted papules; favors Japanese men	Mononuclear infiltrate around acrine ducts within the epidermis and dermis; spongiosis	
Secondary syphilis	Annular plaques with central hypopigmentation; favors face; flu-like symptoms; additional cutaneous manifestations; positive for RPR and FTA-ABS	Perivascular and interstitial infiltrate with plasma cells; also lichenoid or granulomatous infiltrate, exocytosis	Ch. 82
Mycosis fungoides	Annular erythematous or hypopigmented plaques, some with scale; advanced with classic lesions; often pruritic	Infiltrate of atypical lymphocytes in the papillary dermis with epidermotropism	Ch. 120

DIFFERENTIAL DIAGNOSIS OF FIGURATE ERYTHEMA			
Disorders	Clinical features	Histologic findings	Cross references
<i>Lesions with absence or variable presence of scale</i>			
Neutrophilic sebaceous adenitis	Annular, indurated, erythematous plaques that favor the face and upper trunk of young adults; may be exacerbated by sun exposure	Focal necrosis of sebocytes; neutrophils within sebaceous glands; superficial and deep lymphocytic infiltrate	
<i>Pustular lesions</i>			
Pustular psoriasis (annular pattern)	Rapid expansion with erythematous borders studded with pustules	Subcorneal and spongiform pustules; neutrophils in the epidermis	Ch. 8
Sneddon-Wilkinson disease, IgA pemphigus	Pustules in a circinate pattern; peripheral expansion with central clearing; axillae, groin, inframammary areas	Subcorneal pustules; neutrophils in the epidermis; intercellular epidermal IgA deposition in IgA pemphigus	Chs 8 & 29
Eosinophilic pustular folliculitis (Ofuji disease)	Grouped follicular pustules and papulopustules in an annular or serpiginous configuration; favors the face, trunk and arms	Infundibular eosinophilic pustules	Ch. 38
<i>Erosive/vesiculobullous lesions</i>			
Erythema multiforme (EM), Stevens-Johnson syndrome	Association with HSV and <i>Mycoplasma</i> vs. drugs, respectively; target lesions more common in EM; mucosal involvement	Necrotic keratinocytes, vacuolar degeneration of the basal layer and variable inflammation	Ch. 20
Necrolytic migratory erythema	Erosion and crusting of advancing edge; periorificial involvement; glossitis; association with glucagonoma or liver/GI disorders (pseudoglucagonoma); constitutional symptoms	Pallor and edema of the upper epidermis; variable erosion	Ch. 53
Bullous pemphigoid, pemphigoid gestationis	Annular urticarial plaques; tense bullae; pruritus; peripheral blood eosinophilia	Subepidermal bullae with eosinophils > neutrophils; linear deposits of IgG and/or C3 in BMZ	Ch. 30
Linear IgA bullous dermatosis	Annular urticarial plaques; vesicles and bullae within border ("string of pearls"); pruritus	Subepidermal bullae with neutrophils > eosinophils; linear deposits of IgA in BMZ	Ch. 31
Epidermolysis bullosa simplex (especially Dowling-Meara subtype)	"Herpetiform" blisters in figurate array; episodic flares of migratory circinate erythema and blistering; K5 or K14 mutations	Intraepidermal split; eosinophilic inclusions within the keratinocytes	Ch. 32
Annular epidermolytic ichthyosis	Migratory, annular or polycyclic, erythematous plaques with superficial blistering/pooling and scaling at the border; episodic flares; K1 or K10 mutations	Epidermolytic hyperkeratosis	Ch. 57
<i>Purpuric lesions[†]</i>			
Purpura annularis telangiectoides	"Cayenne pepper" petechiae and telangiectasias in annular configuration; favors legs	Capillary dilation and tight perivascular infiltrate of lymphocytes; extravasated erythrocytes; siderophages	Ch. 22
Acute hemorrhagic edema of infancy	Annular or targetoid, edematous, erythematous plaques that become purpuric; favors face, ears and extremities; febrile but non-toxic-appearing child <2 years of age	Leukocytoclastic vasculitis; perivascular IgA deposition in fewer than one-third of cases	Ch. 24
<i>Perforating lesions</i>			
Elastosis perforans serpiginosa	Peripheral edge composed of keratotic papules; favors neck and other flexural areas	Excess of degenerated elastic fibers in clumps in the superficial dermis with transepidermal elimination	Ch. 96

[†]Purpura can also occur in urticarial vasculitis.

